

07b · The same campaign question, on real data — Google’s geo experiment (companion)

July 12, 2026

The business decision. A real advertiser spent **\$50,000** on a real ad campaign, in **50 US geos, for four weeks** (16 Feb – 15 Mar 2015), and left 50 geos dark. This is the worked example published with Google’s geo-experiment methodology (Kerman, Wang & Vaver 2017, the `matched_markets` library) — daily **sales and ad cost per geo**, exactly the panel your own company’s data warehouse holds. The question is the one notebook 07 trained us for: **how much incremental sales did the spend cause, and did it pay for itself?**

0.0.1 What changes when the data is real

Notebook 07 planted a +12% lift in a simulator and graded synthetic control against it. Here nobody tells us the truth — that is the whole epistemic situation of applied causal inference. Three things replace the planted truth:

1. **A randomized benchmark.** The 100 geos were split into treatment and control **at random** (in matched pairs, as we’ll verify from the data itself). So a design-based estimate — the matched-pairs difference-in-differences — is *unbiased by construction*. (Recall 07’s difference-in-differences: the *change* in the treated group, post minus pre, minus the same change in the control group; randomizing which geos are treated makes that difference unbiased for the campaign effect, because the two groups can then differ only by the coin flip.) It is noisy, but it is the closest real data ever gets to a known answer, and we grade the synthetic control against it.
2. **Falsification tests** that need no truth: placebo campaigns on untreated geos (Depth B), a fake launch date, and a check the simulator could never give us for free — the campaign *ends*, so the estimated effect must **switch off when the money does**.
3. **Real messiness:** missing geo-days, heavy-tailed market sizes (the largest geo sells 300x the smallest), and a weekly cycle you can see with the naked eye. Each one bites somewhere below, and one of them (market size) breaks a naive estimator in Depth A.

0.0.2 How this notebook is organised

The same **7-step contract** as every notebook — with step 2’s “simulate a ground truth” honestly replaced by “load real data, and say what can still be validated” — plus the anchor depths:

- **Depth A** — the naive estimators on a *single* real market (where they fail for real), and a convex-hull failure the pre-fit gate catches
- **Depth B** — placebo inference: pseudo-cell permutation, the scale-free RMSE-ratio, placebo-in-time

- **Depth C** — the dollar decision and its sensitivity to the launch date, the donor pool, and the margin

FAST=False SC sampling={'draws': 1500, 'tune': 1500, 'chains': 4} pseudo-cells=200

0.1 2 · Load the real data (what replaces “simulate a ground truth”)

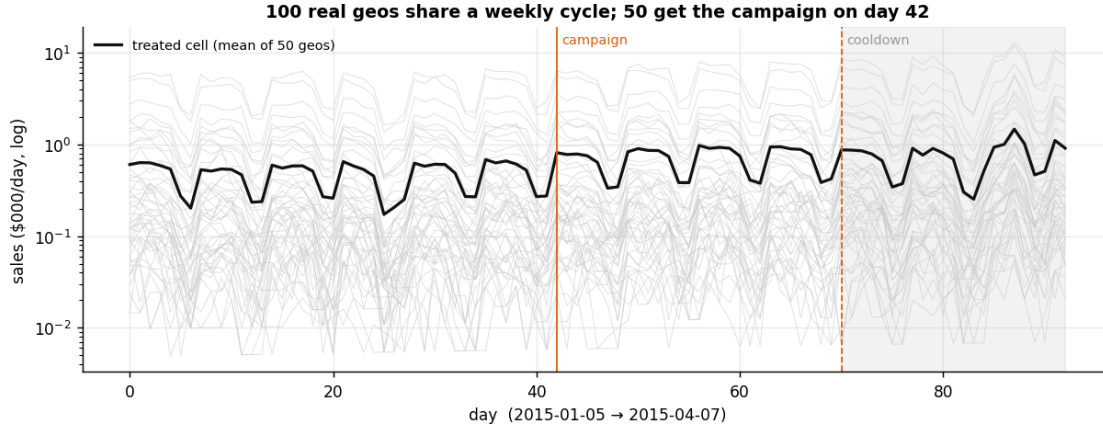
`cmp.data.load_google_geo()` fetches the two public CSVs (Apache-2.0, pinned commit; ~230 KB — cached under `~/.cache/cmp` after the first call) and returns the long panel: `date`, `geo`, `sales`, `cost`, `group`. The **data card**, verified in code below:

fact	value
units	100 US geos, numbered by size (geo 1 = largest — visible in the data)
period design	93 days, 2015-01-05 -> 2015-04-07, daily randomized in matched pairs : consecutive size-ranked geos split 1 treated / 1 control
treatment	ad campaign in the 50 group-2 geos, 16 Feb – 15 Mar (28 days), \$50,000 total spend
cooldown	23 days <i>after</i> the spend stops — a free falsification window
blemishes	75 of 9,300 geo-days missing at source (small geos, mostly pre-period) — we interpolate and disclose

Two working conventions. **(a)** We analyse sales in **\$000s** — same numerical scale the fitters were built for. **(b)** The treated unit is the **test cell**: the *mean* daily sales per treated geo. The mean (not the sum) keeps the treated series inside the donors’ range — a sum of 50 geos could never be a convex combination of single geos (the synthetic counterfactual is only ever a weighted average of the donor geos, so it can never rise above the biggest donor or fall below the smallest — the treated series must sit *inside* the donors’ range, their ‘convex hull’), and Depth A shows what happens when a series escapes the donor hull.

Eyeball the panel below before any model: every geo rides the same weekly cycle (the shared factors of 07, no longer a modelling assumption but a visible fact), and the treated cell visibly jumps *inside the campaign window only*.

100 geos (50 treated / 50 control) · 93 days · campaign days 42-69 (28 days) ·
cooldown 23 days
interpolated geo-days: 75/9300 (0.8%) - small geos, mostly pre-period
ad spend: \$0 before launch (no anticipation), \$50,000 inside the window, \$0
after - the design is clean
geo sizes are heavy-tailed: largest \$6,728/day vs smallest \$18/day (~369x)



0.2 3 · Identify — the estimand, and what randomization buys us

The estimand. Total incremental sales caused by the campaign in the treated geos over the 28-day window:

$$\tau = \sum_{t=T_0}^{T_1} \left(Y_{\text{cell},t}(1) - Y_{\text{cell},t}(0) \right) \times 50,$$

where $Y_{\text{cell},t}$ is mean sales per treated geo and the $\times 50$ restores cell totals. Recall 07’s potential-outcome notation: $Y_{\text{cell},t}(1)$ is sales *with* the campaign (what we observe in the treated geos) and $Y_{\text{cell},t}(0)$ is what those same geos *would* have sold with no campaign — the never-observed counterfactual the synthetic control must reconstruct. The estimator is exactly 07’s: simplex-weighted donors fit to the pre-period fill in the missing $Y_{\text{cell},t}(0)$ (see 07 §3 for the full math and the convex-hull argument). The dollar decision then divides by spend: **iROAS** = $\tau / \$50\text{k}$, incremental revenue per ad dollar.

The assumptions, upgraded by the design. In 07 most rows of this table were unverifiable promises; here randomization converts the worst one into a guarantee:

assumption	status on this real dataset
convex hull / no extrapolation	<i>checked:</i> the treated cell (a mean of 50 geos) sits mid-range of the donors; Depth A shows geo 1 violating it and the gate catching it
good pre-fit	<i>checked:</i> 42 pre-days, PASS/FAIL gate in Step 5
no anticipation	<i>checked from data:</i> \$0 of spend before day 42
stable co-movement	<i>supported:</i> shared weekly cycle visible; placebo-in-time (Depth B) probes it
no spillover between geos	<i>assumed:</i> geos are distinct media markets, but adjacency isn’t in the data — the honest caveat that survives even randomization

assumption	status on this real dataset
no concurrent treated-only shock	guaranteed in expectation by randomization — a shock would have to coincide with a random coin flip across 50 pairs; in 07 this row was “institutional knowledge only”

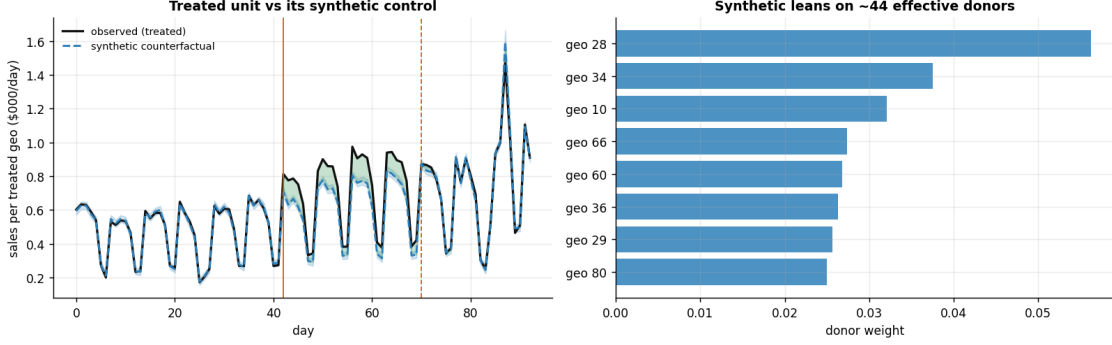
Why bother with synthetic control if the design is randomized? Because the design-based estimate, while unbiased, is *noisy* — geo sizes are so heavy-tailed that a plain treatment-vs-control comparison has a wide interval (Step 5 quantifies this). Synthetic control re-weights donors to *match* the treated cell and cuts that variance. And the moment you leave experiments — one treated market, chosen not by coin flip — SC is the only member of this family still standing. That is the real lecture point of this notebook: **the experiment validates the method we’ll use when there is no experiment.**

0.3 4 · Estimate — Bayesian synthetic control on the test cell

Same model as 07: **Dirichlet-simplex weights** over the 50 control geos, fit to the 42 pre-launch days, returning a posterior over the counterfactual path — and therefore over total lift and iROAS. Read three things off the fit, exactly as in 07: **which donors** carry weight (the Dirichlet spreads it — expect a high “effective number of donors” on a mean-of-50 target), the **pre-fit RMSE** (Step 5 turns it into a PASS/FAIL gate), and the **counterfactual band** (a posterior predictive, carrying the cell’s own daily noise, not just weight uncertainty).

And read the convergence line — including the divergences. The sampler already runs at `target_accept = 0.95`, yet NUTS still reports some divergent transitions (the exact count, r-hat, and ESS are printed below). The geometry is the suspect: 50 simplex weights pressed against a noise scale of ~2% of the level create funnel-like corners the integrator occasionally overshoots. We do *not* wave this away — divergences are the one warning that can invalidate a fit even when r-hat looks perfect. Two checks argue these are benign: the r-hat/ESS readout below says the chains agree on the same posterior (**r-hat** near 1.0 means the independent chains landed on the same distribution; **ESS** — effective sample size — is the number of *effectively independent* draws behind each estimate, and the several hundred printed below is healthy), and — decisive — Step 5 refits the *same* objective with a divergence-free SLSQP optimizer (SLSQP is a plain numerical optimizer that finds the single best simplex weights directly, with no sampling and therefore no divergences, so it solves the *same* fit as a clean cross-check on the Bayesian total) and prints how far the two totals sit apart (a small fraction of the posterior’s own 90% width). If that cross-check disagreed, the right move would be reparameterization, not narration.

```
SC sampling convergence: max r-hat 1.000 - min ESS 3268 - divergences 56
pre-fit RMSE $10/geo/day on a ~$472 level (~2%)
total incremental sales $149k 90% CI [$141k, $158k] on $50k of spend
revenue iROAS 2.98 90% CI [2.81, 3.15]
```



0.4 5 · Validate — against the randomized experiment

Four checks. The first is the one no simulator can fake: an *unbiased, design-based* estimate of the same estimand, from the experiment’s own randomization.

1. **The matched-pairs DiD anchor.** The assignment pattern in the data shows consecutive size-ranked geos were randomized one-treated-one-control (a matched-pairs design — we verify this in code). So the mean of the 50 within-pair difference-in-differences is unbiased for the campaign effect, with an honest standard error from the pair scatter. Concretely, with $\bar{Y}$ the within-window daily means for the treated/control geo of pair p :

$$d_p = (\bar{Y}_{p,\text{post}}^{\text{tr}} - \bar{Y}_{p,\text{pre}}^{\text{tr}}) - (\bar{Y}_{p,\text{post}}^{\text{ct}} - \bar{Y}_{p,\text{pre}}^{\text{ct}}), \quad \hat{\tau}_{\text{DiD}} = \underbrace{50 \times 28}_{\text{geos} \times \text{days}} \cdot \frac{1}{50} \sum_{p=1}^{50} d_p, \quad \widehat{\text{SE}}(\hat{\tau}_{\text{DiD}}) = 50 \times 28 \cdot \frac{s_d}{\sqrt{50}},$$

where s_d is the sample standard deviation of the 50 pair differences — exactly the code’s `pair_diffs / did_total / did_se`. *Why pairing cuts variance:* each d_p compares a geo to its size-matched neighbour, so the size factor differences out *within* every pair; with the heavy right tail of geo sizes (the largest sells ~370x the smallest, Step 2), that tail inflates s_d far less than it would inflate the spread of a plain two-group DiD. The anchor is our stand-in for 07’s planted truth: **the synthetic control point should land inside its interval**. Note how *wide* that interval still is — unbiasedness isn’t precision; this is exactly the variance the model is there to cut.

2. **Pre-fit gate** — same concrete rule as 07: pre-RMSE below a third of the daily lift the design must detect, printed PASS/FAIL.
3. **The off-switch falsification.** The spend stops on day 70; a real advertising effect should collapse toward zero in the 23-day cooldown (up to carryover). A gap that *persists* after the money stops would smell of a confound rather than a campaign. This replaces the simulator’s “we know the truth after week 40” with a physical prediction the data can refute.
4. **Leave-one-out donors** — drop each top-weighted donor, refit (fast SLSQP), watch stability.

What we cannot do on real data — and say so plainly, as 07’s multi-seed check taught us: we cannot refit on fresh copies of the world, so we cannot *measure* interval calibration here. 07 found

SC's total-lift band runs **optimistic** even when the point is right; carry that lesson — trust the placebo permutation (Depth B) for significance, not the band width alone.

anchor: matched-pairs DiD total \$147k +/- 42 (90% CI [78, 216]k; iROAS 2.94 [1.56, 4.32])

SC total \$149k (iROAS 2.98) - lands on the anchor's point with a ~3x tighter band (the variance reduction SC is for; mind 07's optimism lesson on band width)

pre-fit gate: pre-RMSE \$10 vs $|\text{daily lift}|/3 = \$36 \rightarrow \text{PASS}$

off-switch: campaign-window gap \$+107/geo/day $\rightarrow$ cooldown \$-3/geo/day

(collapses when the spend stops - and no long carryover to book as extra value)

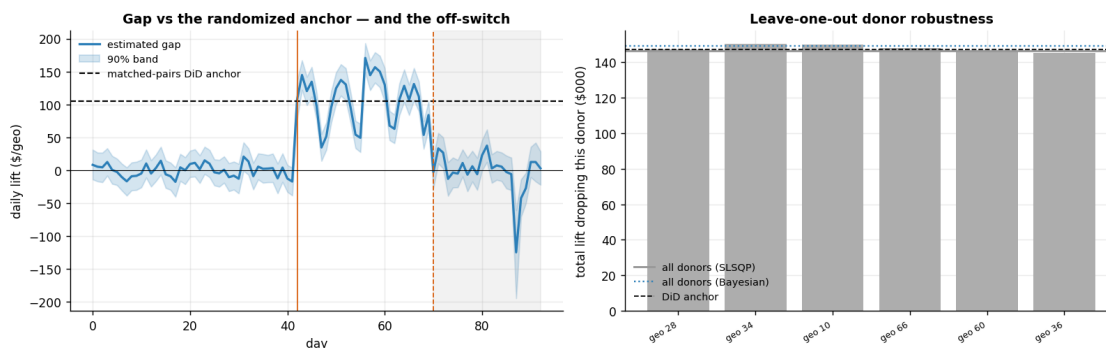
leave-one-out: total ranges \$145-150k vs SLSQP baseline \$147k - no single donor drives it

divergence cross-check: 56 divergent transitions, but the divergence-free SLSQP refit of the same objective lands \$2.5k from the Bayesian total (15% of the 90%-CI width) $\rightarrow$ benign (simplex funnel corners, not bias)

interpolation check (a): dropping the 10 gappy donors (near-uniform Dirichlet weight 19%) moves the daily lift \$+105 $\rightarrow$ \$+105/geo/day (0.2% shift) $\rightarrow$

interpolation is immaterial

interpolation check (b): dropping all 17 gappy geos on both sides gives \$+121/geo/day for the surviving 43-geo cell - pure composition (bigger geos, bigger per-geo lift), matched by the anchor on the same pairs (\$+121); relative lift 22.2% vs 22.3% - unchanged



Read-out. The synthetic control lands almost exactly on the randomized answer — **\$149k vs the anchor's \$147k** (iROAS 2.98 vs 2.94) — passing the only exam real data can set. And the comparison teaches why the model exists at all: the *unbiased* anchor carries a +/- \$70k-wide 90% interval (heavy-tailed geo sizes make even 50 randomized pairs noisy), while SC's band is ~3x tighter around the same point. The gap path shows a clean rectangle: nothing before day 42, a sustained ~\$105/geo/day lift during the four paid weeks, and a collapse to ~-\$3/geo/day within the cooldown — the effect **switches off when the money does**, which simultaneously validates the attribution and tells the media planner there is no adstock annuity to book. Leave-one-out totals move by ~\$3k on a \$147k estimate: no single market carries the result.

0.5 6 · Decide, in dollars

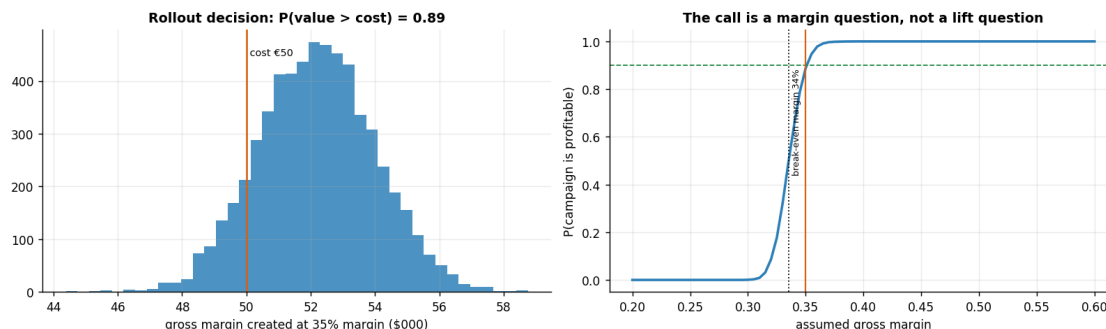
The trap in every iROAS deck: **3.0 revenue-iROAS is not “3x return”**. Ad spend buys *revenue*; the firm keeps only the **gross margin** on it. The campaign pays if

$$\text{iROAS} \times \text{margin} > 1 \quad \Longleftrightarrow \quad \text{iROAS} > \frac{1}{\text{margin}},$$

so at our assumed 35% margin the bar is not 1.0 but **2.86** — and our posterior sits almost on top of it. That is the honest shape of this decision: the *lift* is beyond doubt (Depth B), the *profitability* is a knife-edge that turns entirely on a finance input the model cannot see. The right briefing separates the two.

```
{
  "P_value_gt_cost": 0.889,
  "expected_value": 52.189,
  "expected_net_value": 2.189,
  "expected_roi": 0.044,
  "value_lo": 49.251,
  "value_hi": 55.165,
  "cost": 50.0,
  "decision": "TEST FURTHER"
}
```

break-even margin = $1/\text{iROAS} = 33.5\%$; the campaign clears $P(\text{profitable}) \geq 0.90$ only for margins $\geq 35.5\%$



0.6 7 · Caveats

- **The estimate travels; the decision might not.** iROAS ~ 3 is for *this* creative, *this* spend level, *these* 50 geos, in Feb–Mar. A national rollout extrapolates on all four axes — diminishing returns alone (nb06’s saturation curves) argue the marginal iROAS at 10x spend is lower.
- **Profitability hinges on an input outside the data.** The margin sweep is the honest deliverable; a single “GO” at 35% margin overstates what the sales panel can know.
- **Spillover is the surviving assumption.** Randomization can’t stop Denver’s ads reaching Boulder’s shoppers; if spend leaks into control geos, both the anchor and SC are biased *downward* (control geos absorb some lift), making our estimate mildly conservative.

- **Interval optimism.** 07's multi-seed check showed SC's total-lift band under-covers even when the point is right. The agreement with the anchor's *point* is earned; the band's exact width is not.
- **75 interpolated geo-days** (0.8%, small geos, mostly pre-period) — not asserted away: the interpolation check in Step 5 reruns the fit without the gappy donors (a sub-1% shift, printed) and again with every gappy geo dropped outright on both sides — the per-geo lift moves, but the matched-pairs anchor restricted to the same surviving pairs moves with it and the *relative* lift is unchanged: a composition effect, not an interpolation artifact.
- **One experiment.** This is a single draw of the world, and we've used its randomization once; the multi-seed calibration story lives in 07 where worlds are free.

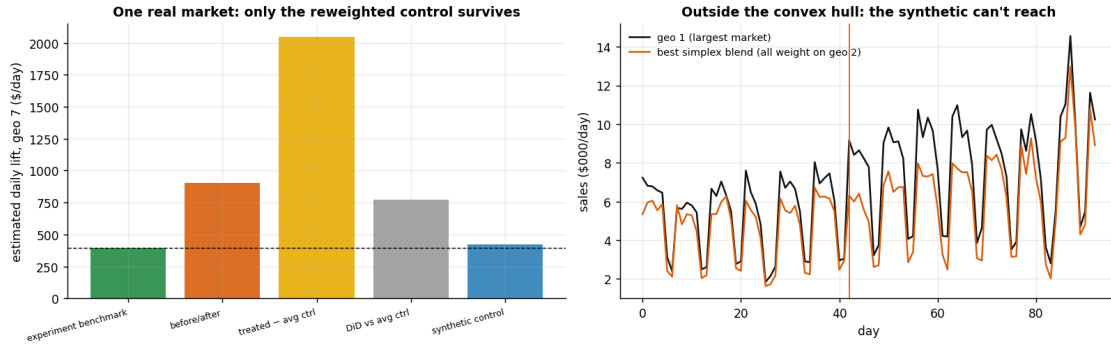
1 Depth A · The naive estimators, on a single real market

In Step 5 the naive comparisons would have looked *fine* — with 50 randomized geos aggregated into a cell, treated-minus-control is unbiased by design; that is precisely why companies run geo experiments. The moment the question becomes 07's — **one** treated market, no coin flip — the shortcuts break on real data exactly as they broke in simulation. We replay the bake-off pretending only **geo 7** (a mid-size market, ~\$1.8k/day) ran the campaign, grading against the benchmark the full experiment implies for it (its share of spend x the anchor iROAS => ~ **+\$394/day**):

- **before/after** — books the season: sales everywhere rose into spring;
- **treated — average control** — books the *level* gap: geo 7 is ~3x the average control's size;
- **DiD vs average control** — fixes the level, keeps the wrong *sensitivity*: a big geo swings more per unit of common shock than the average-of-50, so shared waves don't cancel;
- **synthetic control** — reweights donors to match geo 7's own scale and rhythm.

Then the failure 07 could only warn about: **geo 1 is the largest market in the country** — bigger than *every* control geo, hence **outside the convex hull**. The simplex does its honest best (all weight on geo 2, the largest donor), the pre-fit is visibly bad, and the **pre-fit gate FAILs** — the method itself tells you not to trust its number. That self-diagnosis is worth more than the estimate.

```
{
  "experiment benchmark": 394,
  "before/after": 904,
  "treated \u2212 avg ctrl": 2049,
  "DiD vs avg ctrl": 771,
  "synthetic control": 423
}
geo 7 gate: pre-RMSE $122 vs |daily lift|/3 = $141 -> PASS    (weights spread
over ~5 donors)
geo 1 gate: pre-RMSE $868 vs |daily lift|/3 = $665 -> FAIL  (effective donors
1.0: all weight on geo 2; implied iROAS 5.1 vs the experiment's 2.94 - inflated
~75%)
```



Read-out. Against geo 7's $\sim +\$394/\text{day}$ benchmark: **before/after** lands at **$+\$904$** (2.3x too big — it credits the campaign with spring), **treated — average control** at **$+\$2,049$** (5x — it books geo 7's size as lift), **DiD** at **$+\$771$** (2x — level fixed, sensitivity still wrong: a $\$1.8\text{k}/\text{day}$ market swings harder per common shock than the $\$0.6\text{k}/\text{day}$ average control), and **synthetic control** at **$+\$423$** , **within $\sim 7\%$** , with the gate PASSing (pre-RMSE $\$122$ vs the $\$141$ bar). Same data, four estimates, one survivor — 07's bake-off, no simulator required. And geo 1 is the counterexample that keeps us honest: the country's largest market has no counterfactual among smaller donors, the gate FAILs loudly (pre-RMSE $\$868$ vs a $\$665$ bar), and the estimate it would have produced — iROAS ~ 5.1 vs the experiment's 2.94 — is exactly the number an unscrupulous deck would have shipped. **When the treated unit escapes the donor hull, aggregate or walk away.**

2 Depth B · Placebo inference — is the gap statistically real?

Same inferential logic as 07, adapted to the design. The treated unit is a *cell* — the mean of 50 geos — so the fair placebo is not a single geo but a **pseudo-cell**: draw 25 of the 50 *control* geos at random (the largest cell that still leaves a donor pool), pretend they were a campaign cell, fit them a synthetic control from the remaining 25, and record the post-window “gap.” None of these bundles saw a single ad dollar, so their gaps map out pure noise — **the permutation distribution the design itself generates**. One scale mismatch to own: a mean of 25 geos is noisier than the real cell's mean of 50 (roughly 2x the idiosyncratic variance), and each placebo fit gets only 25 donors instead of 50 — both push the placebo gaps *wider* than the real cell's own null, so the permutation p is, if anything, **conservative**; a real cell that still tops these noisier placebos has cleared a higher bar than the nominal p suggests. Two test statistics, as in 07: the raw mean post-gap, and Abadie's scale-free **post/pre RMSE ratio** (which doesn't reward bundles that simply fit badly everywhere). We also rerun 07's **placebo-in-time** — a fake launch 14 days before the real one, inside the pre-period.

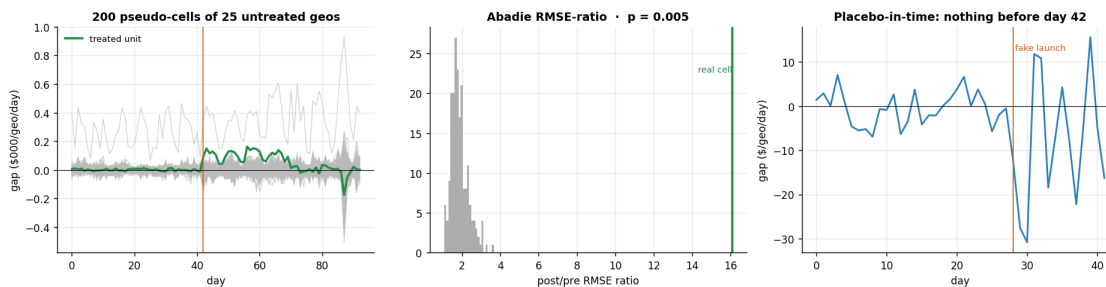
This permutation test is the *calibrated* significance statement for this notebook — the posterior band is the optimistic one (07's multi-seed lesson), the rank among placebos is honest by construction.

pseudo-cell permutation: real gap $\$+105/\text{geo}/\text{day}$ vs placebo 95% range $[-17, +21]$

-> $p = 0.010$

RMSE-ratio test: real cell 16.1 vs placebo median 1.78 -> $p = 0.005$ (no placebo bundle comes close: the real cell ranks 1st of all 201 units)

placebo-in-time: fake-launch gap $\$-8/\text{geo}/\text{day}$ - 7% of the real $\$+107$ effect and the wrong sign (pre-fit RMSE $\$4$) - no anticipation, no invented lift



Read-out. The spaghetti plot says it before any statistic does: the real cell’s post-launch gap ($\sim +\$105/\text{geo}/\text{day}$, printed above) sits far outside anything the untreated pseudo-cells produce, and Abadie’s scale-free RMSE ratio — the real cell near 16 against a placebo median near 2 — removes the objection that it merely fits badly everywhere. Read the printed p-values carefully, though. First, a permutation p has a *floor* of $1/(N_{\text{pseudo}} + 1) - 1/201 \approx 0.005$ at the FULL run’s 200 placebos — so hitting the floor means “the most extreme unit we observed,” not an arbitrarily small tail probability; the print above reports the real cell’s actual rank rather than asserting it. Second, recall the scale mismatch flagged above: these 25-geo placebos are *noisier* than the 50-geo real cell, so the printed p is conservative. The placebo-in-time closes the loop: a fake launch 14 days inside the pre-period finds $\sim -\$8/\text{geo}/\text{day}$ — under a tenth of the real effect, and the wrong sign — no anticipation, no invented lift.

3 Depth C · The dollar decision & sensitivity

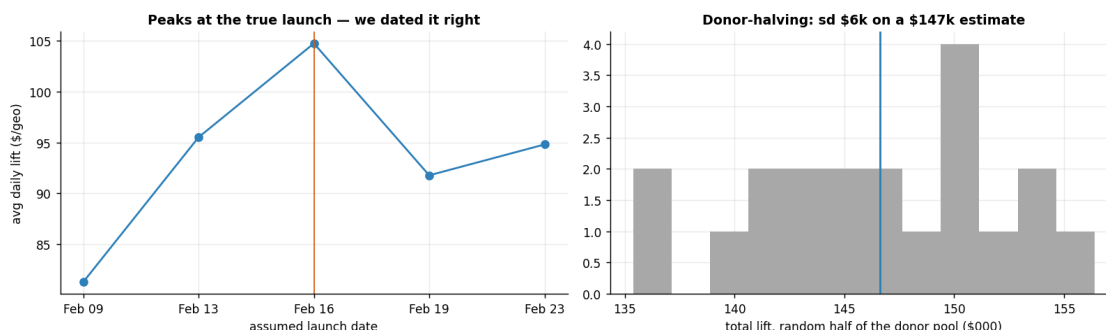
Three stress tests on the number a CMO would repeat in a board meeting, then the verdict. **Launch-date sweep:** as in 07, mis-dating the campaign folds no-effect days into the window and dilutes the average — the estimate should peak *at* the true launch, and the peak is itself a diagnostic that we dated it right. **Donor-pool halving:** refit on random halves of the donor pool; a result that owns its donor pool shouldn’t lurch. **The margin x iROAS surface** is already in Step 6; here we print the summary a decision memo needs.

```
{
  "total_incremental_sales_k": 149.1,
  "ci90_k": [
    140.7,
    157.6
  ],
  "iroas": 2.98,
  "did_anchor_k": 147.2,
  "did_anchor_se_k": 41.9,
  "pre_rmse_gate": "PASS",
  "p_pseudo_gap": 0.01,
```

```

    "p_rmse_ratio": 0.005,
    "placebo_in_time_gap": -7.8,
    "cooldown_gap_per_day": -2.5,
    "break_even_margin": 0.335,
    "P_profitable_at_35pct_margin": 0.889,
    "decision_at_35pct": "TEST FURTHER"
}

```



3.0.1 The one-paragraph decision

The campaign **worked**, and we can prove it to experimental standard: the synthetic control tracks the treated cell to within ~\$10/geo/day before launch (gate PASS), lands on the randomized matched-pairs answer (\$149k vs \$147k of incremental sales; iROAS 2.98 vs 2.94), is the most extreme unit on both placebo permutation tests ($p \sim 0.005$ – 0.010), shows nothing material at a fake launch date (−\$8/geo/day, 7% of the real effect, wrong sign), survives leave-one-out and donor-halving, and — the real-data bonus no simulator provides — **switches off within days of the spend stopping**, which both certifies the attribution and denies any carryover annuity. What the data *cannot* settle is profitability: \$2.98 of revenue per ad dollar clears cost only above a **33.5% gross margin**, and reaches 90% confidence only above ~35.5%. So the briefing is two sentences, not one: *the lift is real and precisely measured; whether to scale is a margin question* — GO for the high-margin lines, NO-GO below one-in-three margins, and any national rollout should expect the marginal iROAS to sag below this test's 2.98 (saturation, notebook 06) — so re-test at scale rather than extrapolate.